Flower Morphology per Genus — Methodology

Zero-shot vision-language classification of 10 SAM3-cropped photos per angiosperm genus | remote_sam3, branch *flower-morphology* | input: all-genera run 2

This document describes how the pipeline in `morphology/` turns run 2's photos into a morphology call per genus: is the flowering structure an inflorescence, a capitulum, a spadix, a grass spikelet? Is the flower actinomorphic or zygomorphic? Every threshold below is the value in the code. Numbers marked **measured** come from data or a test run. Numbers marked **estimated** are projections that the first HiPerGator run will confirm or correct.

1. Approach in one paragraph

No labelled training set of flower morphology exists for these 10,734 genera, so the classifier is **zero-shot**. A general vision-language model (Qwen3-VL-8B-Instruct) is shown each photo with one multiple-choice question at a time. It is not asked to write an answer. A single forward pass gives its probability for the next token, and the probabilities on the option letters (A, B, C...) *are* the answer: a full distribution over the options rather than a single guess. Ten photos per genus are averaged into a genus-level call, with an agreement score that says how consistent the photos were.

Step	What happens	Where
1. Select	10 photos per genus from run 2's metadata	CPU, seconds
2. Crop	Box around each photo's SAM3 masks + margin, letterboxed to 512 × 512	DataLoader workers
3. Score	6 questions × 1 forward pass each → probability per option	1 B200
4. Aggregate	Mean probabilities per genus and family → call, share, agreement, abstain, confidence	CPU
5. Validate	Calls vs. 60 textbook genera (155 trait checks)	CPU
6. Sheets	One contact sheet per genus for checking by eye	CPU job

2. Photo selection

Source: run 2's `all_genera_segment_metadata.csv`, which has 235,262 photos (up to 30 per genus, chosen by phenovision probability tier). Each photo's SAM3 'flower' masks were made at a 0.5 score threshold, keeping the top 5. Per genus, up to 10 photos are taken in this order:

- **Masked photos first.** A photo is masked if it has at least one SAM3 mask and a mask file.
- **One per species before a second from any species**, so a genus is represented by as many of its species as possible.
- Within each round, by **highest mask confidence** (conf_max), then phenovision prediction probability, then photo_id, so the selection is deterministic.
- A genus with fewer than 10 masked photos is filled with its unmasked photos, which are shown uncropped.

Quantity (measured)	Value
Genera	10,734
Photos selected	90,605
... of which have SAM3 masks (cropped)	84,117
Genera with the full 10 photos	8,117
Genera with no masked photo at all (all uncropped)	459

3. Cropping

The model sees the flowers, not the whole scene. For a masked photo:

- Take the union of the photo's kept masks (up to 5) and its bounding box, rescaled to the image if the masks are at a different resolution.
- Grow the box by **25%** of its longer side on every edge to keep context (stems, neighbouring flowers, bracts).
- Never crop smaller than **35%** of the photo's shorter side. A box at the photo's edge slides back inside rather than being clipped, so it keeps its full size.
- Letterbox to **512 × 512** on mid-grey, scaling up or down without distorting the aspect ratio. A fixed size gives every image the same **256 visual tokens** (Qwen3-VL: 16-px patches, 2 × 2 merge = 32 px per token), so batches need no padding.

A photo with no mask, or whose mask file is missing, is letterboxed whole.

4. The questions

Six questions, each with up to 8 options. Options marked *abstain* mean "can't tell from this photo". They are recorded, but they never become a genus's call (§6). The text is in `morphology/questions.py`. Editing it changes the questions and the output columns, and no other code needs to change.

Question	Options (italic = abstain)
unit What is the flowering structure in the photo?	A. **solitary**, B. **inflorescence**, C. **head**, D. **spadix**, E. **spikelet**, F. **catkin**, G. *other*, H. *none*
symmetry What is the symmetry of an individual flower, seen face on?	A. **actinomorphic**, B. **zygomorphic**, C. **asymmetric**, D. *unclear*
display What makes the flowers conspicuous?	A. **petals**, B. **bracts**, C. **stamens**, D. **reduced**, E. *unclear*
fusion Are the petals separate or fused?	A. **free**, B. **fused**, C. *no_petals*, D. *unclear*
form Which best describes the shape of an individual flower?	A. **open**, B. **tubular**, C. **bell**, D. **lipped**, E. **papilionaceous**, F. **orchid**, G. **spurred**, H. *unclear*
merosity How many petals or tepals does one flower have?	A. **3_6**, B. **4_8**, C. **5_10**, D. **many**, E. *unclear*

Every option carries a short description with textbook examples in the prompt (e.g. *spadix*: "a fleshy spike of tiny flowers, usually with a hood- or leaf-like spathe, as in Arum, Anthurium or calla"). The full prompt for the first question:

```
You are an expert botanist. Look at the flowers in this photograph of a wild plant.

What is the flowering structure in the photo?

A. A single flower, or flowers that each stand on their own
B. An inflorescence: a cluster of several distinct flowers (raceme, spike, umbel, panicle,
cyme)
C. A flower head (capitulum): many tiny florets packed together so the whole looks like one
flower, as in a daisy, sunflower or thistle
D. A spadix: a fleshy spike of tiny flowers, usually with a hood- or leaf-like spathe, as in
Arum, Anthurium or calla
```

```
E. Grass- or sedge-like spikelets of chaffy scales (glumes and lemmas), with no petals
F. A catkin: a dangling or upright spike of tiny petal-less flowers, as in willow, birch or
alder
G. Some other structure
H. No flower is visible

Answer with the letter of the best option only.
```

5. Scoring a photo

The prompt is formatted with the model's chat template, with the image first, then the question, then the start of the assistant's turn. One forward pass (bfloat16, SDPA attention, only the last position's logits kept) gives logits z over the whole vocabulary for the first token of the answer. With L the token ids of the option letters (each letter is checked to be a single token):

```
p_k = exp(z_Lk) / sum_j exp(z_Lj) probability of option k
mass = sum_j exp(z_Lj) / sum_v exp(z_v) share of all probability on the letters
```

p is the photo's answer. **mass** is a sanity check. Near 1 means the model understood it was answering a lettered question. A low value means its next token was something else and the letter probabilities should be trusted less. The photo's own call is argmax p over all options, abstain included. photos.csv keeps every option's probability, so nothing is thrown away at this stage.

Why letter probabilities instead of generated text: no parsing, no off-list answers, and no sampling randomness. A whole distribution comes out of one forward pass, and it is cheaper than generation, because nothing is decoded.

6. From photos to a genus call

For a genus with n readable photos and a question with options k, let p_{ik} be photo i's probability for option k and R the non-abstain options:

```
m_k = (1/n) sum_i p_ik mean probability
call c = argmax over k in R of m_k
share = m_c / sum over k in R of m_k
agreement = (1/n) #{ i : argmax over k in R of p_ik = c }
abstain = sum over k not in R of m_k
```

Restricting the call to R matters. Ten photos of tiny flowers would otherwise all vote "can't tell". Instead, the call is the best real option, and *abstain* reports how much the model hedged.

Confidence	Rule (applied in this order)
low	abstain >= 0.5, or agreement < 0.5
high	share >= 0.6 and agreement >= 0.7 and abstain < 0.3
medium	everything else

Each genus also gets a one-line *description* joining its calls, with medium calls marked '?' and low calls left out, e.g. *inflorescence · zygomorphic · petals · fused · lipped?*. Families get the same treatment, with all their genera's photos pooled. genus_probabilities.csv keeps every m_k as features for a later supervised model.

7. Validation

`morphology/reference.csv` lists **60 genera** whose traits are textbook-certain. Only those traits are filled in, so there are **155 checks** in total: unit 28, symmetry 33, display 37, fusion 20, form 24, merosity 13. Examples: *Arum, Arisaema, Anthurium, Zantedeschia, Symplocarpus* → spadix; *Poa, Festuca, Bromus,*

Carex, Cyperus → spikelet, reduced; *Salix, Betula, Alnus, Corylus, Populus* → catkin; *Helianthus, Taraxacum, Bellis, Cirsium, Aster* → head; *Vicia, Lathyrus, Lupinus, Pisum* → zygomorphic, pea-like; *Salvia, Lamium* → fused, two-lipped; *Campanula, Vaccinium, Erica* → fused, bell. Genera where the trait varies within the genus (e.g. wild vs. cultivated *Rosa* petal counts) are left blank for that trait. The summary prints accuracy per question and lists every miss.

Local test (measured). The smaller Qwen3-VL-2B ran on an RTX 4060, on 10 photos each of *Vicia sativa* and *Chrysopsis scabrella* (an aster). It called *Vicia* zygomorphic, petals, pea-like (all high confidence) and *Chrysopsis* a head, actinomorphic. All 3 reference checks were correct. Its weakest question was petal count: it called *Vicia* 3/6 rather than 5, and that question had the lowest letter mass (0.72, against 0.88–0.97 for the others). The 8B model is expected to do better, but petal count should be read with care.

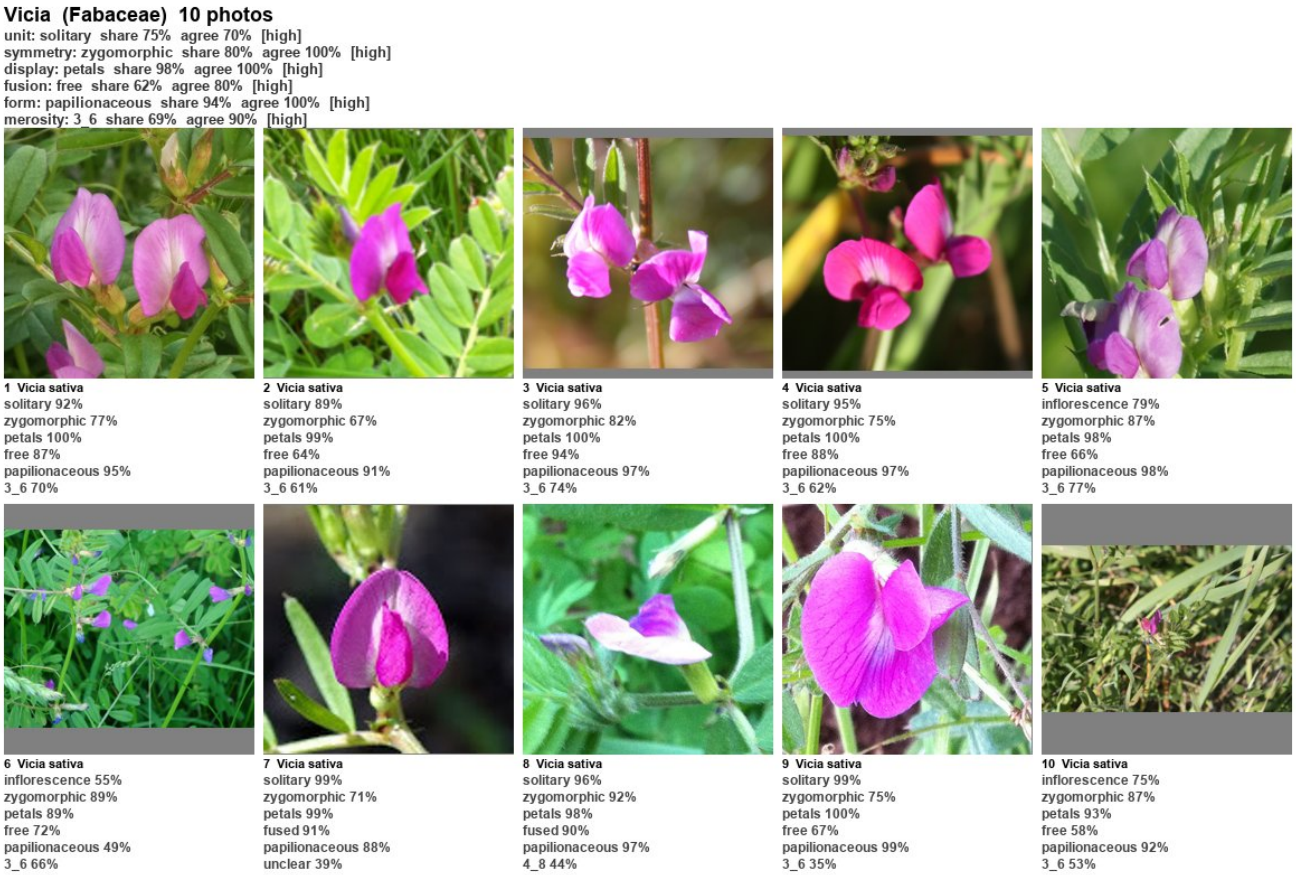


Figure 1. Contact sheet from the local test (Qwen3-VL-2B). Genus calls with share, agreement and confidence on top. Each crop's own calls beneath it. Photo 10 had no mask file in the test and is shown whole.

8. Compute and run time

Quantity	Value	Basis
Photos × questions	90,605 × 6 = 543,630 forward passes	measured
Tokens per pass	~490 (256 image + ~100–240 text)	measured
Tokens per photo, all questions	~2,400	measured
Total prefill tokens	~220 million	derived
2B on RTX 4060	1.56 s per batch of 8 × 492 tokens (~25% of peak)	measured
8B on one B200	~2 hours at similar utilisation	estimated
Job limit	stops itself at 7.5 h, summarises, resubmit resumes	by design

Parallelism. Images are scored in batches of 64 per forward pass. A batch that runs out of memory splits in half and retries. Twelve DataLoader worker processes read the photos and masks from /blue, crop, and run the image preprocessing, once per batch, with 4 batches queued ahead of the GPU. Only the text is tokenised per question, on the main process. On the first batch the job checks that these hand-built inputs equal the processor's own output token for token, and falls back to the processor if they ever differ. One GPU. Each photo is appended to photos.csv as it finishes, so an interrupted job loses nothing.

9. Outputs

File	Contents
genus_morphology.csv	Per genus: call, share, agreement, abstain, confidence per question; description
genus_probabilities.csv	Per genus: mean probability of every option (features for ML)
family_morphology.csv	The same calls per family
photos.csv	Per photo: every option's probability, the photo's call, letter mass, cropped flag
validation.csv	Every reference check: expected, called, share, agreement, correct
sheets/<Family>/<Genus>.jpg	10 crops with per-photo and genus calls

10. Limitations

- **Zero-shot is a first pass, not ground truth.** The model has never been trained on these definitions. Read validation.csv and a sample of sheets before relying on a question.
- **Probabilities are not calibrated.** A share of 0.8 is a relative strength, not an 80% chance of being right. Calibration needs labelled genera.
- **Letter-position bias.** Language models favour some letters. If a question's calls pile up on option A, reorder the options and rerun to compare.
- **The crop decides what is seen.** If SAM3 masked one flower of an umbel, the crop may show a solitary flower. If it masked a whole head, individual florets may be too small to judge symmetry.
- **Ten photos, chosen by mask confidence.** The most confidently segmented photos may favour large, showy, face-on flowers. In a genus whose species differ, the call follows whichever form most of the sampled species share, and agreement drops.
- **Some questions are hierarchical.** Symmetry and petal count are about an individual flower and are often unknowable for heads, spikelets and catkins. Read them as such (their abstain values will be high).

11. Next steps

- Run the 2,000-photo test, then the full run. Read the validation accuracy per question.
- Hand-label a few hundred genera from the sheets, stratified by family, and measure accuracy and calibration properly.
- For questions that matter, train a small supervised classifier (logistic regression or gradient boosting) on genus_probabilities.csv using those labels. Or fine-tune on the image crops if the zero-shot features prove too weak.

Generated by morphology/make_methodology_pdf.py. Thresholds are read from morphology/classify.py and the questions from morphology/questions.py, so this document matches the code it was generated from.